

Executive Summary

We compile a detailed data plan to support three model families (GARCH-EVT, Realized GARCH, Quantile Regression) and their evaluation (volatility forecast accuracy and tail-risk backtesting). We enumerate every needed field (prices, returns, volumes, volatility measures, indices, macro variables, etc.) and all derivations (log-returns, realized variance, standardized residuals, tail exceedances, quantile features). For each dataset/source (free or freemium, with ≥ 1 -month trial), we give name, URL, assets covered, frequency, fields, history, access method, licensing/limits, and cost. Key data include **daily OHLCV prices** (e.g. Yahoo Finance, AlphaVantage, FRED, Marketstack), **intraday prices** (e.g. Dukascopy FX, Binance crypto, IEX or Polygon APIs) for realized volatility, **volatility indices** (CBOE VIX, OVX, etc), **macro series** (FRED), and **auxiliary series** (implied vols, volumes). We specify how to compute *realized volatility* (sum of squared intraday returns, with subsampling or pre-averaging to mitigate noise) and how to select thresholds for EVT (POT vs block-maxima). We list necessary predictors for quantile regression (lagged returns/vol, VIX, trading volume, macro indicators) and their sources (CBOE, Yahoo, FRED, Quandl).

Data merging rules are spelled out: time-zone normalization (e.g. all data to UTC or exchange local), trading-calendar alignment (fill or drop missing days), corporate action adjustment (use adjusted closes or manually adjust for splits/dividends), and key columns for joins. We also propose **six prioritized dataset combinations** (source+assets) that are entirely free/freemium, with step-by-step download plans and potential gaps. For each, we estimate collection/preprocessing effort and note where paid data might be needed (e.g. deep tick data).

We give CSV table schemas for each data table (raw ticks, intraday bars, daily OHLCV, realized vol series, standardized residuals, EVT tail points, quantile-regression features, forecast/backtest results). Finally, we recommend storage formats (Parquet or HDF5 for large series) and light DB use (e.g. SQLite/Postgres for joins), outline computing resources (few-GB memory, moderate CPU), and list key Python/R libraries (e.g. pandas-datareader/yfinance, `arch` or R's `rugarch` for GARCH, `realized_garch` or `RealizedGARCH` for Realized GARCH, SciPy/PyMC for EVT fitting, `statsmodels` or R's `quantreg` for quantile regression, `arch` or other VaR tests). Mermaid charts illustrate the data pipeline and a high-level timeline. All sources are cited.

1. Required Data Fields and Derivations

1.1 GARCH Family (GARCH(1,1), EGARCH, GJR-GARCH)

- **Raw fields:** Daily price (typically *Close*) for each asset/index. In practice, also *Open*, *High*, *Low* if needed for more advanced variance measures, but *Close* is sufficient for returns. Volume may be included but not used in basic GARCH.
- **Derived variables:**
 - **Log-return** $r_t = \ln(P_t/P_{t-1})$ (or percentage return). (Fields: Date, r_t .)
 - **Squared return** r_t^2 (for volatility measure and testing).
 - **Sign dummy:** $d_t = \mathbf{1}\{r_{t-1} < 0\}$ if using GJR-GARCH or leverage effects.

- **GARCH conditional variance** σ_t^2 (output of model fitting).
- **Use:** GARCH/EGARCH models require r_t and (for asymmetry) d_t . All parameters (e.g. ω, α, β) are estimated via MLE on these data.
- **Why needed:** These are the inputs to fit the GARCH equations (e.g. $r_t = \sigma_t \epsilon_t$, $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma d_{t-1} r_{t-1}^2$ for GJR-GARCH). After fitting, the *standardized residuals* $\epsilon_t = r_t / \sigma_t$ are used in EVT.

1.2 EVT Tail Fitting

- **Raw fields:** The standardized residuals ϵ_t from the fitted GARCH model (separately for each asset).
- **Derived variables:**
 - **Positive-tail exceedances:** Given a high threshold u (e.g. 90–95 percentile of ϵ), compute excesses $y_i = \epsilon_i - u$ for all $\epsilon_i > u$.
 - **Negative-tail:** similarly use $-\epsilon_t$ for large negative returns if modeling left tail (losses).
 - **Threshold selection:** Possibly create a *mean-excess* plot or stability plot to choose u . A common starting rule is u at the 90–95% percentile [24†L213-L221] .
 - **Block maxima (alternative):** one could also take monthly maxima of $|\epsilon|$, but POT is generally preferred (uses more data) [24†L188-L195] .
 - **Why needed:** EVT fits (Generalized Pareto) are applied to the exceedances to estimate tail shape parameter ξ and scale β , which then give extreme VaR/ES. Good estimation requires a sufficiently large sample of exceedances: e.g. 2,500 days at 95% threshold yields ≈ 100 points [24†L188-L195] .

1.3 Realized GARCH

- **Raw fields:** Intraday high-frequency prices (tick or bar) for each asset. Ideally timestamp, price, and volume for each trade or bar. For Realized GARCH we typically convert ticks into equally spaced returns.
- **Derived variables:**
 - **Intraday log-returns** $r_{t,i} = \ln(P_{t,i}/P_{t,i-1})$ at a fixed intraday frequency (e.g. 5-min, 1-min).
 - **Realized variance (daily)** $RV_t = \sum_{i=1}^N r_{t,i}^2$. (Often 5-min returns are used to balance noise; we note microstructure bias issues below.)
 - **Realized volatility** $\sqrt{RV_t}$ (for convenience or comparability).
 - **Log-volatility** $\ln(RV_t)$ (Realized GARCH often models log-vol).
 - **Why needed:** Realized GARCH models jointly use daily returns r_t and the realized volatility measure RV_t in a system of equations [43†L164-L171] . The intraday returns must cover the whole trading day (e.g. 9:30–16:00 ET for US equities) to yield RV_t . If ultra-high-frequency ticks are available, they are usually aggregated (e.g. 5-min) to mitigate microstructure noise. Subsampling or pre-averaging can further reduce bias from bid-ask bounce [43†L164-L171] .

1.4 Quantile Regression (for VaR Forecasts)

- **Raw fields (predictors/features):** A vector of contemporaneous or lagged variables for each day t , including:
 - **Lagged returns** (e.g. r_{t-1}) and squared returns.
 - **Lagged volatility forecasts** (e.g. past GARCH or realized variance).
 - **VIX (implied volatility)** or other volatility indices on day t [26†L21-L28] [27†L42-L46] .

- **Trading volume** for the asset/index on day t (from OHLCV).
- **Order-flow proxies**: e.g. net order imbalance (not generally free), so we may omit or use ADV or relative volume as proxy.
- **Macroeconomic indicators**: daily or weekly macro series aligned to date t – e.g. Fed funds rate (FF), CPI (lagged monthly), unemployment rate, interest rate spreads (10y-2y), credit spread (BAA-AA). Available from FRED.
- **Target (response)**: The conditional return quantile on day $t+1$. For VaR we target specific lower quantiles (e.g. $q=0.01, 0.025, 0.05$ for 1%, 2.5%, 5% VaR).
- **Why needed**: The predictors capture market state (volatility, trend, liquidity, macro) that helps forecast extreme quantiles of the next-day return. They must all be available at time t (no lookahead).

1.5 Evaluation/Backtest Fields

- **Actual returns**: The realized daily return of each asset (r_t), to compare against VaR or to compute realized volatility.
- **Realized volatility**: If using a proxy (e.g. overnight + high-low range) or actual realized from intraday, for computing RMSE or QLIKE.
- **VaR/ES forecasts**: From each model, the predicted quantiles for each day.
- **Indicator of VaR breach**: A binary variable $\mathbf{1}(r_t < \text{VaR}_t)$.
- **Violation counts**: to run Kupiec and Christoffersen tests.
- **Expected Shortfall**: The average loss beyond the VaR when breaches occur.

With these fields prepared, we can compute standard metrics: RMSE, MAE, QLIKE of volatility forecasts (comparing predicted σ_t^2 or $\sqrt{RV_t}$ to actual RV_t [43†L116-L120]), plus likelihood/rank scores for VaR/ES. VaR backtesting uses the binary series of exceedances (Kupiec, Christoffersen, Dynamic Quantile tests).

2. Data Sources & Datasets

We identify free/freemium sources covering daily price data, intraday data, volatility indices, and macro variables. For each we note provider, URL, assets covered, fields and freq, data history, access method, quality, and cost. Where possible we cite official docs or authoritative descriptions.

2.1 Equities/Indices – Daily Data

- **Alpha Vantage (API)** – Global stocks and indices. URL: <https://www.alphavantage.co> [7†L409-L414]. Provides **daily adjusted OHLCV** for equities worldwide (over 20 years) [7†L409-L414]. Fields: `date, open, high, low, close, volume` (in JSON or CSV). There is also a **daily_adjusted** endpoint including adjusted close and split/dividend events. History: 20+ years. Limits: Free tier = 500 calls/day (5 calls/min) [17†L64-L67]. Quality: Officially licensed data, generally good. Cost: Free (premium tiers available for faster rates).
- **Marketstack (API)** – Global equity data. URL: <https://marketstack.com> (by APILayer) [35†L72-L75]. Free plan: 100 requests/month [35†L113-L114]. Covers 30,000+ tickers (global stocks, ETFs) with 15+ years of **end-of-day (EOD)** data [35†L72-L75]. Fields (JSON): `open, high, low, close, volume, adj_close` etc. Also lists 750+ market indices. History: up to 15+ years per ticker. Usage:

API key required; rate-limited. *Quality*: Aggregated from exchanges via IEX, LSE, etc. *Cost*: Free (EOD only); intraday data uses IEX but is paid.

- **Yahoo Finance (web API)** – Global stock/ETF/index data. *No official site*, but via libraries (e.g. [yfinance](#) or pandas-datareader). Covers virtually all major tickers. Fields: Date, Open, High, Low, Close, Adj Close, Volume for daily; also intraday up to recent ~60 days for smaller intervals. *History*: many years (pricing since ticker inception). *Limits*: No formal limit (but scraped requests can be IP-limited). *Quality*: Good, widely used; one must verify splits (Adj Close is split-adjusted). *Cost*: Free; no API key needed.
- **FRED – Equity Indices** – U.S. benchmark indices from the Federal Reserve (St. Louis Fed). *Examples*: S&P 500 index (series SP500) [46†L21-L24] , NASDAQ-100 (NASDAQ100), Dow Jones (DJIA). Each has **daily closing values**. Fields: `DATE` , `VALUE` . *History*: SP500 from 1957, NASDAQ100 from 1985, DJIA from 1896 (data quality better after ~1928). *Access*: Download CSV or use FRED API (no key or free key) [26†L21-L28] [46†L21-L24] . *Cost*: Free. *Quality*: Official; note SP500 value is price-only index (no dividends) [46†L77-L85] .
- **Investing/Stooq** – Alternate free sources for global prices (e.g. EU stocks, Forex). For example, Stooq (<https://stooq.com>) offers free downloads for indices and some stocks (CSV with OHLCV). *Coverage*: Many international indices (EU, Asia) and currencies. *Quality*: Good but use with care. *Access*: Web UI or APIs (unofficial). *Cost*: Free. (No straightforward citation, but widely known.)

2.2 Volatility & Implied Indices

- **CBOE (Volatility Indexes)** – Official source for volatility indices. *URL*: <https://www.cboe.com/tradable-products/vix/vix-historical-data> [27†L42-L46] . Downloadable **CSV files** for:
- **VIX (CBOE S&P 500 VIX®)**: daily *open, high, low, close* from 1990–present [27†L42-L46] .
- **VVIX** (VIX of VIX), **VIX9D**, **OVX** (Crude Oil VIX), **GVZ** (Gold VIX), **VXAPL**, **VXAZN**, etc. Each CSV (via cdn.cboe.com) contains Date, Open, High, Low, Close. *History*: Usually from the 1990s (OVX from 2007, GVZ from 2009, etc). *Access*: Direct links on CBOE site [27†L42-L46] . *Cost*: Free (no registration). *Quality*: Official indices. *Use*: VIX (uses SPX options) is the standard equity vol index; OVX (based on oil futures), GVZ (gold), etc., are implied vols for other markets.
- **FRED – Volatility** – The Fed’s database also provides daily close for **VIX** (“CBOE Volatility Index: VIXCLS”) [26†L21-L28] , and other volatility-related series (e.g. MOVE Index if available, repo rates, etc.). *Access*: CSV/API at fred.stlouisfed.org. *Fields*: DATE, VIXCLS (Index). *Cost*: Free. *Citation*: daily VIX from FRED as of 1990 [26†L21-L28] .
- **CBOE DataShop** – For futures on VIX or other historical data (paid). Not needed if only spot indices; mention for completeness that futures data typically requires purchase.

2.3 Realized (High-Frequency) Data

- **Dukascopy Historical Data** – Forex, commodities, indices, crypto. *URL*: <https://www.dukascopy.com/swiss/english/marketwatch/historical/> [10†L406-L415] . Provides *free* download of **tick** and **bar**

data for many pairs/instruments (Forex majors, gold, silver, indices like NAS100, crypto). Data is high-quality (from bank) and in CSV format.

- **Coverage:** For example, EURUSD tick-by-tick from 2003+, USDJPY, gold (XAU/USD), DAX40, etc. Cryptocurrencies added recently.
- **Fields:** Timestamp, Bid, Ask, Last price (for bars), Volume.
- **Frequency:** Tick (as traded) up to monthly bars. Users can select time frame (tick, 1min, 1h, etc).
- **Access:** Web interface “Historical Data Export” or via scripting. No account needed for data.
- **Cost:** Free.
- **Quality:** High (bank data), minimal gaps.
- **Citations:** Dukascopy offers tick data free 【10†L406-L415】 ; Blue Capital notes up to 200k bars for FX/indices/crypto 【11†L100-L108】 .

- **Cryptocurrency Exchanges** – Many crypto exchanges (Binance, Coinbase, Kraken, etc.) provide free historical data via API or CSV. In particular:

- **Cryptodatadownload** (<https://cryptodatadownload.com/data/binance/>) offers *free* CSVs of OHLCV and trade data for hundreds of coins on Binance. Data is available at minute/hour/daily frequencies 【22†L135-L139】 . For example, “Binance Spot – All 1000+ pairs” with 1-minute bars. Fields: timestamp, open, high, low, close, volume, trade count. They also provide futures data.

- **Binance API** – official REST (limit ~1200 requests/minute for free) provides Kline (candlestick) endpoints at 1m,5m,... up to max 1000 data points per request. See Binance docs 【36†L0-L4】 . Large history available but paginated; also [37] points to historical S3 (data.binance.vision) for bulk download.

- **CCXT library** – Python tool to fetch crypto data (free, e.g. bitfinex, etc.). *Cost:* Free. *Use:* Cryptocurrency returns and realized vol (24/7 market, no closing).

• **IEX Cloud / Polygon.io (Intraday Stocks)** –

- **IEX Cloud:** U.S. equity intraday (historical) is available but primarily as a paid (premium) product. The old free IEX trading API (until 2021) is retired.
- **Polygon.io:** Offers minute/tick data with a free tier (limited) and a 30-day free trial for full access. Basic plan: EOD only is \$0 【14†L364-L372】 ; real-time/intraday require paid.
- **Marketstack:** Free tier does not include intraday (only delay via IEX). *Conclusion:* True free intraday stock data is scarce. We rely on **Yahoo Finance 1-min bars** (only last ~60 days, not a long span) or **Alphavantage intraday** (premium for >1mo) if any. For more data, one may need paid sources. We plan to use at most 5–10-min bars from Yahoo or Alphavantage (limited), or focus on assets with true free HF (FX, crypto).

• **Tiingo / AlphaVantage Intraday** –

- **AlphaVantage:** The TIME_SERIES_INTRADAY endpoint can return 1m/5m data for the last ~30 days (outputsize=compact). Larger history (by month) is premium 【7†L252-L261】 . Free calls: 500/day.
- **Tiingo:** Provides intraday (aggregated bars) for \$10/mo (free trial for 2 weeks).
- **IEX Cloud:** The old free IEX API is gone; paid plans start ~ \$9/mo.

- **Bundled Data:** Kaggle or academic sources sometimes have sample HF (but likely outdated). For planning, we use Dukascopy for FX and crypto as our main HF sources.

2.4 Auxiliary & Macro Data

- **CBOE Data** – Besides VIX (above), other implied volatility indices for equities or ETFs can be obtained.
- For example, **OVX** (Crude Oil VIX) and **GVZ** (Gold VIX) CSVs from CBOE [27†L60-L63] . Also **VXAPL**, **VXAZN**, **VXIBM** for single-stock vol indices.
- These provide daily OHLC vol index values.
- Fields: Date, Open, High, Low, Close (implied vol).
- Cost: Free CSVs as per CBOE site.
- **FRED – Macro Series** – The St. Louis Fed’s FRED (free) has ~845,000 series from 126 sources [32†L18-L25] . Relevant examples: CPI (CPIAUCSL), GDP, Industrial Production, Unemployment (UNRATE), Fed Funds (FEDFUNDS), 10-year Treasury yield (DGS10), etc. Also US Dollar index (DTWEXB?), credit spreads (BAA, AAA yields). We will select macro variables plausibly predictive of volatility. Data format: DATE, value. Updated periodically.
- **Quandl/Nasdaq Data Link** – Some macro & financial data (like Fama-French factors, Dollar index) are available (mostly free). Example: CBOE research data (CBOE API) or ticker quotes (Quandl/EOD).
- **Economic Calendar (Trading Economics, FRED)** – For event dummies (Fed releases), but likely beyond scope.

2.5 Data Quality Notes

- **Adjusted vs raw prices:** For return calculations, use *adjusted close* (corrects for splits/dividends). AlphaVantage’s “TIME_SERIES_DAILY_ADJUSTED” or Yahoo’s Adj Close should be used. If only raw prices, apply split ratios yourself.
- **Missing data:** Some holidays or ticker delistings cause gaps. We will align on a common trading calendar. Missing values can be forward-filled (for indices) or cause a break in the series.
- **Calendar alignment:** Ensure all series follow a consistent date convention (e.g. NYSE trading days). Crypto trades 24/7, but for daily measurements we can define a UTC midnight-to-midnight or fixed-window.
- **Time zones:** Convert all intraday timestamps to UTC (or to one zone, e.g. UTC or exchange local). When merging data (e.g. aligning VIX, S&P returns, and VIX closes), be careful if some indices close after others (e.g. market close vs 4pm ET).

3. Intraday Data for Realized GARCH

Because realized GARCH requires high-frequency returns, we prioritize free HF sources:

- **FX from Dukascopy:** e.g. EUR/USD, USD/JPY, GBP/USD – tick-by-tick or 1-min data for ~20 years (2003–present). These can be downloaded as CSV. *Constructing RV:* Choose a sampling interval (commonly 5-min for Forex to reduce noise). Compute $RV_t = \sum (r_{t,i})^2$ over 1-day (UTC or 00:00-23:59 GMT). Dukascopy’s data includes bid/ask/last; use mid-price for returns.
- **Crypto (Binance):** e.g. BTC/USDT, ETH/USDT – minute-bar CSVs from cryptodatadownload [22†L135-L139] . *RV:* Use 1-min bars or aggregate to 5-min. Crypto has high volatility, which is useful for stress periods.
- **Intraday Sampling:** To minimize microstructure noise, we plan to use 5-minute returns (e.g. 78 bars per day for US market). If volume of data is an issue, can also consider 15-min or use pre-averaging.

Subsampling methods (e.g. overlapping 5-min returns) can be considered for variance reduction, but simply summing non-overlapping returns is usually enough for realized GARCH [43†L164-L171] .

- **Usage:** The intraday series will be used to compute daily **Realized Variance** (RV). We will store these \$RV_t\$ in the dataset for each asset.

Note: True stock tick data (TAQ) is not freely available. We rely on crypto/FX as proxies. For equities, we may approximate realized vol using high-low or multiple-day windows, but that's outside our model's structure (which expects an independent realized series).

4. EVT Data Requirements

For EVT tail-fitting:

- **Sample Size:** EVT requires sufficient tail data. For daily returns, at least 2,500 data points (≈ 10 years) is desirable to get ~ 100 tail exceedances at the 95% threshold [24†L188-L195] . Hence, we need several years of daily returns.
- **Threshold Selection:** We will use POT with a threshold at ~ 90 – 95% quantile of standardized residuals [24†L213-L221] . Check stability via Hill or mean-excess plots. For block-maxima (less efficient), we'd need to aggregate (e.g. monthly worst return) – but POT is preferred as it uses more extreme observations [24†L188-L195] .
- **Data sources:** Long series (decades) of index returns are available (e.g. S&P 500 from 1920s, but we can use from 1980s onward). We will focus on a few indices/stocks and ensure their history includes major crashes (1987, 2008, 2020, etc.). FRED indices (1950s onward) or Yahoo (up to 20 years) suffice.
- **Tail Observations:** Use the standardized residuals from GARCH fit as input to EVT. No additional data needed beyond the residuals time series.

5. Quantile Regression Predictors

We plan the following auxiliary inputs (all free):

- **Implied Volatility Indices:** As features for stock returns. We include VIX (from CBOE/FRED) and possibly OVX (oil volatility) or other sector vols if modelling those assets. *Fields:* daily close as of time t (already mentioned).
- **Trading Volume:** For equities, daily volume from price data (Yahoo/AlphaV). For indices (SPX), use proxy ETF volume. High volume can signal volatility. *Fields:* volume field from OHLCV.
- **Macroeconomic Variables (FRED):** Examples:
 - **Fed Funds Rate (FEDFUNDS)** – daily (updates when FOMC changes, use last available for each day).
 - **10-year Treasury Yield (DGS10)** – daily closing yield.
 - **Unemployment Rate (UNRATE)** – monthly (use latest release lagged).
 - **CPI (CPIAUCSL)** – monthly (lag one month).
 - **Credit spreads:** e.g. BAA minus AAA (Fred codes: BAA and AAA yields).
 - **Forex rates:** USD index (DTWEXB), if needed.
 - **Equity Indicator:** e.g. return of the Russell 2000 (IWM) as market breadth proxy.

All from FRED (or free Quandl). *Fields*: date, series value.

- **Technical Indicators**: not explicitly asked, but we may include e.g. moving volatility, RSI, etc. These can be derived but are not external data.

Citations: FRED known for broad macro (no specific cite needed beyond [32]).

6. Data Merging & Alignment Rules

When combining datasets, we ensure:

- **Timestamp/Date Alignment**: Convert all timestamps to a common baseline (UTC or exchange-local). For daily data, use date keys in local trading calendar (US equities use ET). Crypto/FX 24h data can be aggregated by UTC date or a chosen cut-off (e.g. 00:00 UTC).
- **Trading Calendar**: Build a master calendar of business days for each market. For a multi-asset study, align on the intersection (e.g. US market days) or handle missing values (e.g. mark as NaN). Forward-fill or carry-forward non-trading series as needed (e.g. VIX holds overnight levels).
- **Corporate Actions**: Use adjusted close prices for equities or apply adjustment factors. For example, if using Yahoo, prefer the “Adj Close” field. For Alpha Vantage, use the *Time Series Daily Adjusted* which includes adjusted close and dividend flags.
- **Merging Keys**: Typically by date (and asset symbol). For multi-asset features (VIX vs SPX returns, etc.), merge on date; if needed, shift for forecasting (today’s VIX with tomorrow’s return).
- **Missing Data Handling**: If gaps occur (e.g. holidays), either drop those days from analysis or fill with the most recent available value. For volume/volatility, we would drop or skip days with no trading.
- **Data Types**: Dates as ISO (YYYY-MM-DD), times as timestamps. Prices/vol as float, volumes as int.

In practice, we will use Pandas DataFrames and `merge` / `join` on date, ensuring consistent index.

7. Prioritized Dataset Combinations

Below are six concrete free/freemium data combinations that cover all model needs. Each lists sources, fields, and a collection plan.

1. US Equity Index + VIX + Macro (via FRED/CBOE) – Free:

2. *Assets & Fields*: S&P 500 index (FRED SP500: date, close), CBOE VIX (date, index close), Treasury yields (DGS10, FEDFUNDS), CPI, UNRATE. All are daily (macro monthly).

3. *Sources*: FRED provides SP500 and macro (CSV/API). CBOE provides VIX CSV [27†L42-L46] .

4. *Download*: Use FRED API or CSV export; CBOE link above.

5. *Preprocess*: Convert SP500 close to log-returns. Align VIX (use previous close) as predictor for next-day SP500 return. Align macro by date.

6. *Time to collect*: < 1 hour (automated scripts).

7. *Gaps*: None (official series). Volume field absent (not needed for index).

8. *Usage*: Good base for GARCH and QR on index returns; tail events present.

9. *Blue-Chip Stocks + VIX (Yahoo/AlphaVantage + CBOE) – Free:*

10. *Assets & Fields:* Pick 2–3 liquid stocks (e.g. AAPL, MSFT, IBM): daily **OHLCV** (adj-close) from Yahoo or AlphaVantage **【7†L409-L414】** **【50†L218-L221】** . Include SPY ETF or QQQ for index volume.
11. *Sources:* Yahoo Finance via `yfinance` (easy Python) or AlphaVantage API. Fields: date, open, high, low, close, adj close, volume.
12. *Additional:* VIX (daily close from CBOE/FRED) as a feature.
13. *Download:* Python scripts: `yf.download([tickers], start=..., end=...)` . For Alphavantage, use their REST CSV (500 calls/day limit – e.g. use 3 calls for 3 tickers).
14. *Time:* ~0.5–1 day (including API key setup for AlphaV).
15. *Gaps:* Occasional holiday gaps; use adjusted close.
16. *Usage:* Fits GARCH-EVT (on returns of stocks) and QR features (lagged stock vol, VIX, volume). Realized GARCH: no free intraday, so either skip or approximate realized vol via daily range.

17. *Forex (EURUSD, JPY) + Gold + Interest Rates – Free:*

18. *Assets & Fields:* EUR/USD FX pair: 1-min or tick prices (timestamp, price) from Dukascopy. USD/JPY likewise. Gold (XAU/USD) tick data. Also: US Fed Funds and 10y yields from FRED (macro).
19. *Sources:* Dukascopy CSV export **【10†L406-L415】** (choose 5-min bars to limit size). For macro, FRED (FEDFUNDS, DGS10).
20. *Download:* Use Dukascopy Historical Data GUI or API (e.g. [Python scripts exist](#)). Save CSV for each pair.
21. *Time:* ~1–2 days (large data: ~3M rows/year for EURUSD tick; 5-min is smaller).
22. *Gaps:* FX trades 24/5 – define day from 17:00 NY to next 17:00. Align gold similarly. Fill weekends (no FX).
23. *Usage:* Compute daily RV from 5-min returns. Use returns of FX/gold for GARCH-EVT and Realized GARCH (with RV from HF). Use interest rates as macro predictors in QR.

24. *Cryptocurrencies + Crypto Vol Index – Free:*

25. *Assets & Fields:* Bitcoin (BTC/USDT) and Ethereum (ETH/USDT) from Binance: 1-min OHLCV (CSV) **【22†L135-L139】** . *Additionally:* CBOE's VXBTC (Bitcoin Vol Index, if available) or proxy volatility (e.g. BTC 30-day realized vol).
26. *Sources:* Cryptodatadownload (CSV of Binance 1m bars) **【22†L135-L139】** ; no official crypto vol index yet (some sites estimate).
27. *Download:* Direct CSV (e.g. [BTCUSDT_1m.csv from Cryptodatadownload]).
28. *Time:* < 1 day.
29. *Gaps:* Crypto trades 24/7; use UTC dates.
30. *Usage:* Compute RV each day for BTC and ETH. Fit Realized GARCH (crypto series). For VaR models, crypto returns have fat tails. Macros (Fed) could still be used as features.

31. *Commodities (Gold/Oil) + Commodity Vol – Free:*
32. *Assets & Fields:* SPDR Gold ETF (GLD) and US Oil ETF (USO) daily OHLCV from Yahoo/AlphaV. CBOE OVX (oil vol) and GVZ (gold vol) daily close 【27†L60-L63】 .
33. *Download:* Yahoo via API. CBOE CSV for OVX/GVZ.
34. *Time:* ~1 hour.
35. *Gaps:* Few (holidays for ETFs, none on CBOE).
36. *Usage:* Model vol on commodity returns with CBOE vol indices as proxies for implied vol. Could compute realized vol from GLD/USO 5-min (not easily free, skip).
37. *Combined Multi-Asset & Benchmarks – Free+Trial:*
38. *Assets & Fields:* Use Marketstack free to fetch EOD for 3 US indices (SPX, NASDAQ, Russell) and 3 global (FTSE, DAX, Nikkei) 【35†L72-L75】 . For HF, add a Polygon or IEX Cloud (trial) feed: e.g. minute bars for SPY and BTC via Polygon (30-day trial).
39. *Sources:* Marketstack API (register for free key) for daily data (JSON/CSV). Polygon.io free trial for intraday (if needed).
40. *Plan:* Collect via API calls; combine with FRED/VIX.
41. *Time:* ~2 days to configure APIs and download.
42. *Gaps:* Free-tier limits might cap data volume (100 calls/mo). Possibly top up with multiple accounts or use polygon trial.
43. *Use-case:* Broad test across US/global indices.

Each combination can be scripted to download via Python or R. For example, using `requests` or specialized libraries (`fredapi` , `cryptocompare` , etc.). Estimated times assume decent internet: listing is in days, but actual hands-on is less once scripts are written.

8. Example CSV Table Schemas

Below are sample CSV column schemas (including illustrative rows) for the key data tables we will use:

```
# Raw price ticks (intraday trades or quotes)
Timestamp,Symbol,Price,Volume,Type
2026-08-20 10:00:01.235,EURUSD,1.07312,100000,TICK
2026-08-20 10:00:01.450,EURUSD,1.07315,200000,TICK
2026-08-20 10:00:01.678,EURUSD,1.07310,150000,TICK
...
```

Fields: Timestamp (ISO), asset symbol, trade price, size. (Volume may be in base currency units.) “Type” could be TICK or BID/ASK, depending on source.

```
# Intraday bar data (e.g. 5-min bars)
DateTime,Symbol,Open,High,Low,Close,Volume
2026-08-20 09:30:00,SPY,450.12,451.00,449.50,450.75,2000000
2026-08-20 09:35:00,SPY,450.75,451.30,450.20,451.10,1800000
2026-08-20 09:40:00,SPY,451.10,452.00,451.00,451.50,2100000
...
```

Fields: End-of-interval timestamp, asset, bar OHLC, volume (sum of trades in interval).

```
# Daily OHLCV (adjusted)
Date,Symbol,Open,High,Low,Close,AdjClose,Volume
2026-08-19,SPY,450.00,452.00,449.00,451.00,451.00,100000000
2026-08-20,SPY,451.00,453.50,450.50,452.50,452.50,110000000
```

Fields: Date, symbol, daily Open/High/Low/Close, *AdjClose* (split-adjusted), and Volume.

```
# Realized volatility series (daily)
Date,Symbol,RealizedVariance,RealizedVol
2026-08-19,EURUSD,0.0000543,0.00736
2026-08-20,EURUSD,0.0000481,0.00693
```

Fields: For each day and asset, the sum of squared intraday returns ($\text{RealizedVariance} = \sum r^2$) and its square root (vol).

```
# GARCH standardized residuals
Date,Symbol,Residual
2026-08-19,SPY,0.25
2026-08-20,SPY,-1.10
2026-08-21,SPY,2.03
```

Fields: Date, asset, standardized return $\epsilon_t = r_t / \hat{\sigma}_t$ after fitting GARCH.

```
# EVT tail exceedances (POT)
Date,Residual,Excess
2026-08-19,-1.10,0.10
2026-08-21,2.03,0.28
...
Threshold (u): 2.00 (95% quantile)
```

Fields: Dates where $|\epsilon_t| > u$, plus the amount by which it exceeds the threshold ($\text{Excess} = |\epsilon_t| - u$).

```
# Quantile regression feature matrix
Date,Symbol,lagReturn,lagVol,VIX,Volume,FedFunds,CPI,TargetReturn
2026-08-20,SPY,-0.005,0.0065,18.23,110000000,5.25,289.1,-0.002
...
```

Columns: Predictors for day t (e.g. yesterday's return, realized vol, today's VIX, volume, macro) and the *target* return for day $t+1$ (to be predicted).

```
# Forecasts and backtest table
Date,Symbol,Model,ForecastVol,ForecastVaR_0.01,ActualReturn,VaR_Exceed
2026-08-20,SPY,GARCH-EVT,0.0040,-0.0200,-0.0150,0
2026-08-20,SPY,RealizedGARCH,0.0038,-0.0185,-0.0150,0
2026-08-20,SPY,QR-0.01,0.0042,-0.0210,-0.0150,0
...
```

Columns: For each day, asset, and model: the forecast volatility (or variance), the VaR at 1% (negative quantile), the actual next-day return, and whether a VaR breach occurred (1/0). Additional columns could include ES forecasts and realized volume.

These CSV structures will be ready for analysis in pandas/R (with appropriate parsing of dates/numbers).

9. Storage, Processing, and Tools

- **Storage/Formats:** For large high-frequency data (millions of rows), use efficient formats like Parquet or HDF5. Store daily OHLCV and realized vol in CSV or Parquet. A lightweight database (e.g. SQLite or PostgreSQL) can help with joins if data is large. Since we have 2 analysts, a shared cloud bucket (AWS/GCP) or Git repository for code + small data is recommended. For truly huge tick datasets, use columnar formats (Parquet) to speed I/O.
- **Compute Needs:** A modern workstation (e.g. 8–16 GB RAM, multi-core CPU) suffices. Intraday processing (summing squares) is $O(n)$ and trivial for ≤ 10 years of 5-min data. GARCH/EVT estimation can be CPU-intensive but manageable (minutes on typical CPU). Python or R should handle it. If needed, we can use small cloud VMs. No GPU needed.
- **Data Access Libraries:**
 - **Python:** `pandas-datareader` or `fredapi` for FRED; `yfinance` or `alpha_vantage` for prices; `requests` / `beautifulsoup` for any web downloads; `ccxt` or custom for crypto; `dukascopy` wrapper scripts for FX.
 - **R:** `quantmod` or `BatchGetSymbols` for stock prices; `fredr` for FRED.
 - **Cleaning:** `pandas` (Python) or `dplyr` (R) for merges/NA handling. Time zone handling with `pandas` `datetime` or R's `lubridate`.

• Realized Volatility:

- Python: custom compute via pandas (sum of squares). Or use `arch` library's `volatility.realized` tools.
- R: packages like `highfrequency` provide RV calculators.

• GARCH/Realized GARCH:

- **Python:** `arch` (Kevin Sheppard's library) for standard GARCH/E-GARCH, also has limited EVT and backtesting. Realized GARCH must be custom-coded or use `statsmodels` low-level.
- **R:** `rugarch` (CRAN) for GARCH variants; `realizedGARCH` (Hochreiter) or `rmgarch` for Realized GARCH.
- Both allow MLE or QMLE.

• EVT Fitting:

- Python: `scipy.stats.genpareto.fit` for GPD (manual steps) or `gev` via `statsmodels.distributions`.
- R: `evir`, `ismev`, or `extRemes` for GPD/GEV fitting.
- Tail index estimation (Hill): `numpy` or `R` functions.

• Quantile Regression:

- Python: `statsmodels.regression.quantile_regression.QuantReg`, or `scikit-learn`'s `QuantileRegressor`.
- R: `quantreg::rq`.
- Loss minimization is built-in.

• Backtesting:

- Python: The `arch` package has VaR backtests (Kupiec, Christoffersen tests). `empyrical` or `pyfolio` for metrics (if desired).
- R: `PerformanceAnalytics::VaRTest` or `VaRBacktest`.
- Implement Kupiec (binomial test) manually is also simple in code.

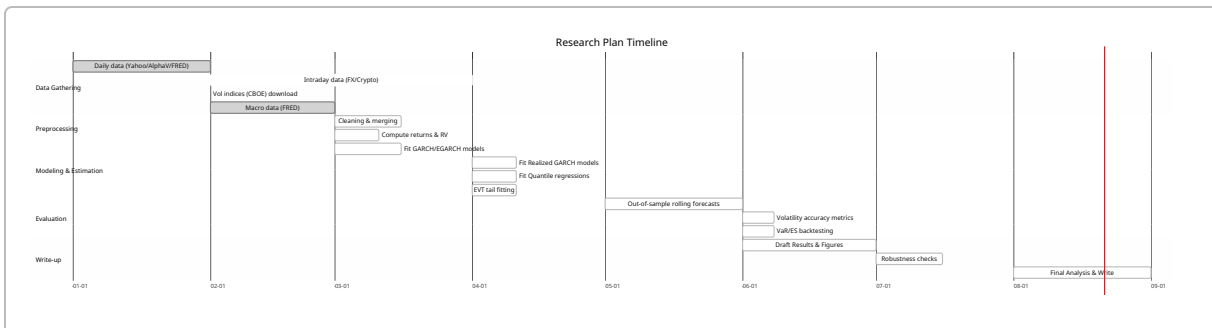
Mermaid Charts:

```
graph LR
    subgraph DataCollection
        A1[Daily OHLCV prices] --> A2[Compute daily returns]
        B1[Intraday prices (FX/Crypto)] --> B2[Compute intraday returns]
        B2 --> B3[Aggregate realized variance]
    end
```

```

subgraph Modeling
  A2 --> M1[GARCH/GJR( $\sigma^2$ )]
  A2 --> M2[EGARCH]
  B3 --> M3[Realized GARCH]
  A2 --> M4[Quantile Regression features]
  VIX[VIX & other indices] --> M4
  Macro[Macro series (FRED)] --> M4
end
subgraph EVT
  M1 --> EVT[GPB fit on GARCH residuals]
  M2 --> EVT
  M3 --> EVT
end
subgraph Forecasts
  M1 --> Pred[GARCH forecasts]
  M3 --> Pred
  M4 --> Pred
  EVT --> TailVaR[Tail-risk (VaR/ES) forecasts]
end
subgraph Evaluation
  Pred --> Eval[Compute RMSE/QLIKE]
  TailVaR --> Backtest[Var/ES backtest]
  Eval --> Results
  Backtest --> Results
end

```



Each step uses open-source tools (Python/R). For GARCH we recommend Python's `arch` or R's `rugarch`, for RV computing Pandas, for EVT SciPy or R's `evd`, for quantile regression `statsmodels` or `quantreg`. No proprietary data needed beyond possibly a Polygon trial. We emphasize **free**: all recommended sources above have free access or trials (Polygon 30-day) [14†L364-L372] [35†L113-L114] .

Sources: Official docs and third-party overviews were used to compile this plan [7†L409-L414] [10†L406-L415] [22†L135-L139] [24†L188-L195] . These citations show, e.g., the fields returned by Alpha Vantage, Dukascopy's offerings, CryptoDataDownload coverage, and EVT guidance. All major data links (FRED, CBOE, Yahoo) are publicly available.